



Van Allen Probe ceases operations

One of the two Van Allen Probes, known as spacecraft B, received a shutdown command. <https://digit.in/aug19-07>



College spending habits imply more

A study revealed your college money habits will ultimately affect when you start seeing yourself as an adult. <https://digit.in/aug19-08>

Cryopegs

Microbes living in isolated packets of brine beneath the Arctic permafrost offers a glimpse into the potential habitats for life in the ice moons of the Solar System.

Aditya Madanapalle | aditya@digit.in

In the 1960s, an organisation called CRREL (Cold Regions Research and Engineering Laboratory), that is part of the US Army, excavated a field site in Northern Alaska. The site was near the city of Utqiagvik. It is the northernmost city in the US, and lies inside the Arctic Circle. The purpose of the excavation was to study wedges of freshwater ice beneath the Arctic permafrost. In the 2000s, researchers collected subsurface brine that was trapped beneath the ice wedges. This water was in a liquid state, because of the sheer amount of salt dissolved in it.

Jody Deming, professor of oceanography at UW, who studies microbial Arctic life explains, “The extreme conditions here are not just the below-zero temperatures, but also the very high salt concentrations. One hundred and forty parts per thousand — 14% — is a lot of salt. In canned goods that would stop microbes from doing anything. So there can be a preconceived notion that very high salt should not enable active life.” Despite the extreme salinity, the water was found to be teeming with living microbes, despite being isolated from the rest of the world for tens of thousands of years. Samples collected in 2017 and 2018 showed thriving colonies of bacteria and viruses. The most common bacteria found is *Marinobacter*. The bacteria, trapped in the saline deposit before being isolated, was a form of marine bacteria. The



The narrow tunnel with pipes to extract samples from the cryopegs

discovery surprised scientists studying brine deposits. Zachary Cooper, a doctoral student said, “We were quite startled at how dense the bacterial communities are.”

The trapped brine deposits are known as “cryopegs”. The temperatures within the cryopegs are between negative 9-to-11 degrees Celsius. Cryopegs are the only habitats on Earth where the temperatures are permanently below freezing, the environment is isolated from any external influence, and has a high amount of salinity. A surprising diversity of life forms are found within

cryopegs, including fungi, yeast and bacteria. The life forms here either thrive because of the high salinity, or have evolved mechanisms to tolerate it. There are both oxygen-breathing aerobic life, and ancient anaerobic life that does not need oxygen to survive. Considering the cold, the metabolic rates of these organisms are very slow. As such, these life forms can be considered extremophiles. Researchers from the University of Washington have been studying the microbial life that is found in these cryopegs. Recently, researchers presented DNA sequences of

Credit: Zac Cooper/UW